Single-qubit gates and measurements

① Unitary ②

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bit $\{0,1\} = \mathbb{Z}_2$

$|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$

Logic gates

in  out

in	out
0	1
1	0

NOT gate — Rev

$UU^\dagger = I$

$U^\dagger = U^{-1}$

in  out

in	out
00	0
01	0
10	0
11	1

AND gate

in  out

in	out
00	0
01	1
10	1
11	1

OR gate

$\left.\begin{matrix}\text{AND gate}\\ \text{OR gate}\end{matrix}\right\}$ Non Rev

video

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Representing a single-qubit system (RECALL)

Pure

ket

State vectors in $\mathscr{H}_2$**:** $|\psi\rangle = \left[\cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}\, e^{i\phi}\,|1\rangle\right] \cancel{e^{i\gamma}}$
where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$.

Operators on $\mathscr{H}_2$**:** $A = \sum_{k=0}^{3} \gamma_k\, \sigma_k$ where $\gamma_k = \mathrm{tr}(\sigma_k\, A) \in \mathbb{C}$.

$[\![\sigma_k, A]\!] = tr(\sigma_k^{\dagger} A)$

Vector-basis	Operator-basis
$\mathcal{B}_2 = \{\|0\rangle, \|1\rangle\}$	$\mathfrak{B} = \{\underset{I}{\sigma_0}, \underline{\sigma_1, \sigma_2, \sigma_3}\} = \{I, \underbrace{X, Y, Z}_{\vec{\sigma}}\}$

Single qubit state: $\rho = \dfrac{I + \vec{r}\cdot\vec{\sigma}}{2}$ where $|\vec{r}\,| \leq 1$.

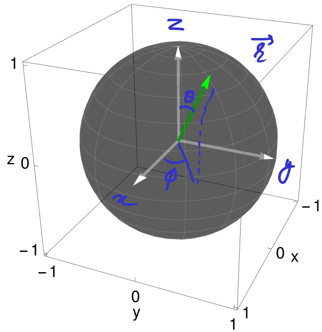
$= |\psi\rangle\langle\psi|$

$0 \leq \rho$

Bloch vector: $\vec{r} = (\langle X\rangle, \langle Y\rangle, \langle Z\rangle) \in \mathbb{R}^3$

$\langle X\rangle = tr(X\rho)$ Born Rule

Bloch Ball



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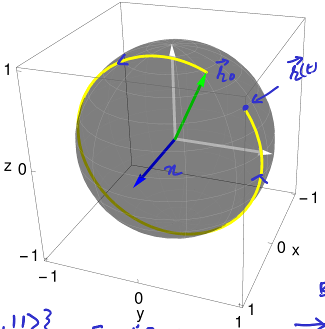
Rotation around x, y, z axes

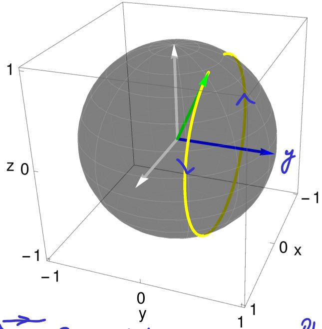
$R_\eta(t) = \cos\left(\frac{t}{2}\right)I - i\sin\left(\frac{t}{2}\right)\hat{\eta}\cdot\vec{\sigma}$

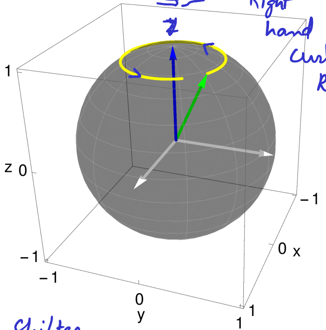
Right hand curl Rule

$t \in \mathbb{R}$

$t = \pi$







$B = \{|0\rangle, |1\rangle\}$

$= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = -iX$

$$R_x(t) = \begin{pmatrix} \cos\frac{t}{2} & -i\sin\frac{t}{2} \\ -i\sin\frac{t}{2} & \cos\frac{t}{2} \end{pmatrix}$$

BS

Beam Splitter

$$R_y(t) = \begin{pmatrix} \cos\frac{t}{2} & -\sin\frac{t}{2} \\ \sin\frac{t}{2} & \cos\frac{t}{2} \end{pmatrix}$$

Phase Shifter

$R_z(t)|0\rangle = |0\rangle$

$R_z(t)|1\rangle = e^{it}|1\rangle$

$$R_z(t) = \begin{pmatrix} e^{-i\frac{t}{2}} & 0 \\ 0 & e^{i\frac{t}{2}} \end{pmatrix}$$

$= e^{-it/2}\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & e^{it} \end{bmatrix}}_{\text{Phase gate}}$

video

$$\vec{r}(t) = (\cos t)\,\vec{r}_0 + (\sin t)\,\hat{\eta}\times\vec{r}_0 + (1-\cos t)(\hat{\eta}\cdot\vec{r}_0)\,\hat{\eta}$$

$$= (\langle X\rangle, \langle Y\rangle, \langle Z\rangle) \quad \text{where } \langle X\rangle = \langle\psi(t)|X|\psi(t)\rangle \text{ and } |\psi(t)\rangle = R_\eta(t)|\psi_0\rangle$$

$y(t)$ $z(t)$ $x(t) = \mathrm{tr}(X\rho(t))$ $\rho(t) = R\rho_0 R^\dagger$

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Unitary evolution: Rotation of the Bloch vector

For a qubit

$U = \exp(-i Ht/\hbar)$ Hamiltonian $H = H^\dagger$ $\frac{H}{\hbar} = \frac{\hat{n}\cdot\vec{\sigma}}{2}$

$\vec{B} \propto \hat{n}$

$\vec{\mu} \propto \vec{S} = \frac{\hbar}{2}\vec{\sigma}$

$H = -\vec{\mu}\cdot\vec{B} = \frac{\omega\hbar}{2}\hat{\eta}\cdot\vec{\sigma}$

Larmor precession $H = \frac{\hbar}{2}\hat{\eta}\cdot\vec{\sigma}$ $\omega = 1$

NMR
Quantum dot

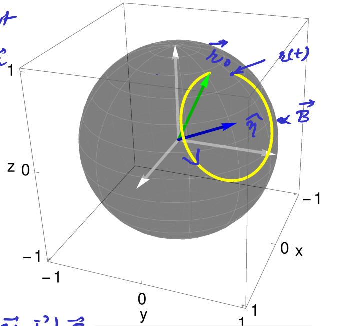
$\langle \vec{S} \rangle = \frac{\hbar}{2}\langle \vec{\sigma} \rangle = \frac{\hbar}{2}\vec{r}_1$

- Rotation of the Bloch vector around the axis $\hat{\eta}$ by an angle t:

$$R_\eta(t) = \exp\left(-\mathrm{i}\,\tfrac{t}{2}\,\hat{\eta}\cdot\vec{\sigma}\right) = \cos(\tfrac{t}{2})\, I - \mathrm{i}\sin(\tfrac{t}{2})\,\hat{\eta}\cdot\vec{\sigma}$$

$$= \mathrm{e}^{-\mathrm{i}\frac{t}{2}}\left(\frac{I + \hat{\eta}\cdot\vec{\sigma}}{2}\right) + \mathrm{e}^{\mathrm{i}\frac{t}{2}}\left(\frac{I - \hat{\eta}\cdot\vec{\sigma}}{2}\right)$$

$f(\hat{\eta}\cdot\vec{\sigma}) = f(+1)\left[\frac{I + \hat{\eta}\cdot\vec{\sigma}}{2}\right] + f(-1)\left[\frac{I - \hat{\eta}\cdot\vec{\sigma}}{2}\right]$



Evolution of state: $\rho(t) = R\,\underbrace{\frac{I + \vec{r}_0\cdot\vec{\sigma}}{2}}_{\rho_0 = |\psi_0\rangle\langle\psi_0|}\,R^\dagger$

$|\psi(t)\rangle = R|\psi_0\rangle$ $|\psi(t)\rangle\langle\psi(t)|$

$RR^\dagger = I$

$$= \frac{I + R\,(\vec{r}_0\cdot\vec{\sigma})\,R^\dagger}{2}$$

$(\vec{a}\cdot\vec{\sigma})(\vec{b}\cdot\vec{\sigma}) = (\vec{a}\cdot\vec{b})I + i(\vec{a}\times\vec{b})\cdot\vec{\sigma}$

$[\cos t/_2\, I - i\sin t/_2\, \hat{n}\cdot\vec{\sigma}]\,(\vec{r}_0\cdot\vec{\sigma})\,[\cos t/_2\, I + i\sin t/_2\, \hat{n}\cdot\vec{\sigma}]$

$$R\,(\vec{r}_0\cdot\vec{\sigma})\,R^\dagger = \boxed{\underbrace{\left[(\cos t)\,\vec{r}_0 + (\sin t)\,\hat{\eta}\times\vec{r}_0 + (1-\cos t)(\hat{\eta}\cdot\vec{r}_0)\,\hat{\eta}\right]}_{\vec{r}(t)}}\cdot\vec{\sigma}$$

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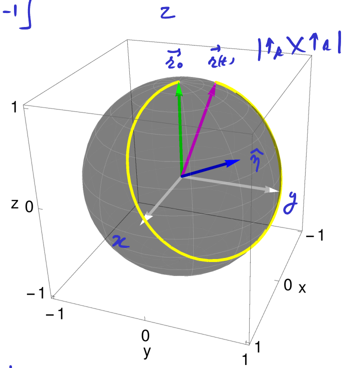
Unitary evolution: Rotation of the Bloch vector

$\hat{\eta} = (\eta_x, \eta_y, \eta_z)$ $\quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$$R_\eta(t) = \cos(\tfrac{t}{2})\, I - \mathrm{i}\sin(\tfrac{t}{2})\, \hat{\eta}\cdot\vec{\sigma}$$

$$= \begin{pmatrix} \cos\frac{t}{2} - \mathrm{i}\sin\frac{t}{2}\,\eta_z & -\mathrm{i}\sin\frac{t}{2}(\eta_x - \mathrm{i}\eta_y) \\ \underbrace{-\mathrm{i}\sin\frac{t}{2}(\eta_x - \mathrm{i}\eta_y)}_{|\uparrow_\eta\rangle = R|0\rangle} & \underbrace{\cos\frac{t}{2} + \mathrm{i}\sin\frac{t}{2}\,\eta_z}_{|\downarrow_\eta\rangle = R|1\rangle} \end{pmatrix} \quad \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{|0\rangle} \quad \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{|1\rangle}$$

in the eigenbasis $\mathcal{B} = \{|0\rangle, |1\rangle\}$ of Z.



$$R\,(\vec{r}_0\cdot\vec{\sigma})\,R^\dagger = \boxed{\overbrace{\left[(\cos t)\,\vec{r}_0 + (\sin t)\,\hat{\eta}\times\vec{r}_0 + (1-\cos t)(\hat{\eta}\cdot\vec{r}_0)\,\hat{\eta}\right]}^{\vec{R}}}\cdot\vec{\sigma}$$

where $\vec{r}_0 = (0,0,1)$

$$R\,[\overbrace{\vec{r}_0\cdot\vec{\sigma}}^{Z}]\,R^\dagger = R\Big[(+1)\overbrace{\left[\frac{I+\vec{r}_0\cdot\vec{\sigma}}{2}\right]}^{|0\rangle\langle 0|} + (-1)\overbrace{\left[\frac{I-\vec{r}_0\cdot\vec{\sigma}}{2}\right]}^{|1\rangle\langle 1|}\Big]R^\dagger$$

$$= (+1)\underbrace{\left[\frac{I+\vec{r}(t)\cdot\vec{\sigma}}{2}\right]}_{|\uparrow_\eta\rangle\langle\uparrow_\eta|} + (-1)\underbrace{\left[\frac{I-\vec{r}(t)\cdot\vec{\sigma}}{2}\right]}_{|\downarrow_\eta\rangle\langle\downarrow_\eta|}$$

$$= \vec{r}(t)\cdot\vec{\sigma}$$

video